Exercise 2

Convex Functions (1 pt)

Simple functions (0.5 pt)

Are the following functions convex? Prove your statement.

1. $f(x) = x^2$, $x \in \mathbb{R}$
2. $f(x) = e^{x^2}$, $x \in \mathbb{R}$
3. $f(x, y) = x^2 + 3xy + 2y^2$, $x \in \mathbb{R}$, $y \in \mathbb{R}$

1.

$$\begin{aligned} & f(x) = x^2 \\ \implies & f'(x) = 2x \\ \implies & f''(x) = 2 > 0 \quad \forall x \in \mathbb{R} \\ & \therefore f \text{ is convex} \end{aligned}$$

2.

$$\begin{aligned} & f(x) = e^{x^2} \\ \implies & f'(x) = 2xe^{x^2} \\ \implies & f''(x) = 2e^{x^2} + (2x)e^{x^2}(2x) = 2e^{x^2}(2x^2 + 1) > 0 \quad \forall x \in \mathbb{R} \\ & \therefore f \text{ is convex} \end{aligned}$$

3.

$$\begin{aligned} & f(x, y) = x^2 + 3xy + 2y^2 \\ \implies & \nabla f(x, y) = (2x + 3y, 3x + 4y) \\ \implies & \nabla^2 f(x, y) = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix} \end{aligned}$$

$\nabla^2 f(x, y)$ is not positive semi-definite since its eigenvalues are $3 + \sqrt{10}$ and $3 - \sqrt{10} < 0$. Therefore f is not convex.

Log-sum-exp (0.5 pt)

Show that the function

$$f(\boldsymbol{x}) = \log(e^{x_1} + \ldots + e^{x_n})$$

is convex on $\mathbb{R}^n$.

Hint: The proof is outlined in Boyd's book p. 74. Develop it yourself and provide the intermediate steps.

Solution: We will develop the proof in Boyd's book.

Let's define for convenience two column vectors:

$$z = \begin{bmatrix} e^{x_1} & e^{x_2} & \ldots & e^{x_n} \end{bmatrix}^T$$

$$\mathbf{1} = \begin{bmatrix} 1 & 1 & \ldots & 1 \end{bmatrix}^T$$

This allows us to rewrite $f(\boldsymbol{x})$ in a more compact manner since:

$$\mathbf{1}^T z = \sum_{i=1}^{n} e^{x_i} \tag{1}$$

$$f(\boldsymbol{x}) = \log(\sum_{i=1}^{n} e^{x_i}) = log(\mathbf{1}^T z) \tag{2}$$

First calculate the gradient:

$$\frac{\partial f}{\partial x_i} = \frac{e^{x_i}}{\sum_{i=1}^{n} e^{x_i}} \tag{3}$$

$$\implies \nabla f(\boldsymbol{x}) = \frac{z}{\mathbf{1}^T z} \tag{4}$$

Then calculate the Hessian:

$$\frac{\partial f}{\partial x_i^2} = \frac{(-1)e^{x_i}}{\left(\sum_{i=1}^{n} e^{x_i}\right)^2} + \frac{e^{x_i}}{\sum_{i=1}^{n} e^{x_i}} \tag{5}$$

$$= \frac{1}{\left(\sum_{i=1}^{n} e^{x_i}\right)^2} \left(\left(\sum_{i=1}^{n} e^{x_i} \right) e^{x_i} - (e^{x_i})^2 \right) \tag{6}$$

$$\frac{\partial f}{\partial x_i \partial x_j} = \frac{(-1)e^{x_i} e^{x_j}}{\left(\sum_{i=1}^{n} e^{x_i}\right)^2} \tag{7}$$

Notice that the outer product zz^T gives us the matrix:

$$zz^T = \begin{bmatrix} e^{x_1} \\ e^{x_2} \\ \dots \\ e^{x_n} \end{bmatrix} \begin{bmatrix} e^{x_1} & e^{x_2} & \dots & e^{x_n} \end{bmatrix} = \begin{bmatrix} (e^{x_1})^2 & e^{x_1}e^{x_2} & e^{x_1}e^{x_3} & \dots & e^{x_1}e^{x_n} \\ e^{x_1}e^{x_2} & (e^{x_2})^2 & e^{x_2}e^{x_3} & \dots & \vdots \\ e^{x_1}e^{x_3} & e^{x_2}e^{x_3} & (e^{x_3})^2 & \dots & \vdots \\ \vdots & \dots & \dots & \dots & \vdots \\ e^{x_1}e^{x_n} & \dots & \dots & \dots & (e^{x_n})^2 \end{bmatrix} \tag{8}$$

This accounts for the terms $-(e^{x_i})^2$ in $\frac{\partial f}{\partial x_i^2}$ and $(-1)e^{x_i}e^{x_j}$ in $\frac{\partial f}{\partial x_i \partial x_j}$.

Next, define a matrix with z in its diagonal:

$$\boldsymbol{diag}(z) = \begin{bmatrix} e^{x_1} & 0 & 0 & \dots & 0 \\ 0 & e^{x_2} & 0 & \dots & \vdots \\ 0 & 0 & e^{x_3} & \dots & \vdots \\ \vdots & \dots & \dots & \dots & \vdots \\ 0 & \dots & \dots & \dots & e^{x_n} \end{bmatrix} \tag{9}$$

This will take care of the terms e^{x_i} in $\frac{\partial f}{\partial x_i^2}$.

Combining the previous results, we can express the gradient with compact notation as:

$$\nabla^2 f(\boldsymbol{x}) = \frac{1}{(\mathbf{1}^T z)^2} \left((\mathbf{1}^T z) \boldsymbol{diag}(z) - zz^T \right) \tag{10}$$

To verify that $\nabla^2 f(\boldsymbol{x}) \succeq 0$ we must show that $\forall v, v^T \nabla^2 f(\boldsymbol{x}) v \geq 0$. Distributing the v and v^T to the inside, we get:

$$v^T \nabla^2 f(\boldsymbol{x}) v = \frac{1}{(\mathbf{1}^T z)^2} \left((\mathbf{1}^T z) v^T \boldsymbol{diag}(z) v - v^T zz^T v \right) \tag{11}$$

We care only about the two terms inside the parentheses:

$$\begin{aligned} v^T zz^T v &= (v^T z)(z^T v) && (12) \\ &= \left(\begin{bmatrix} v_1 & v_2 & \dots & v_n \end{bmatrix} \begin{bmatrix} e^{x_1} \\ e^{x_2} \\ \dots \\ e^{x_n} \end{bmatrix} \right) \left(\begin{bmatrix} e^{x_1} & e^{x_2} & \dots & e^{x_n} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \dots \\ v_n \end{bmatrix} \right) && (13) \\ &= \left(\sum_{i=1}^{n} v_i e^{x_i} \right) \left(\sum_{i=1}^{n} v_i e^{x_i} \right) && (14) \\ &= \left(\sum_{i=1}^{n} v_i e^{x_i} \right)^2 && (15) \end{aligned}$$

And for the other term:

$$v^T \boldsymbol{diag}(z) v = \sum_{i=1}^{n} v_i^2 e^{x_i} \tag{16}$$

Replacing these results we get:

$$v^T \nabla^2 f(\boldsymbol{x}) v = \frac{1}{(\mathbf{1}^T z)^2} \left(\left(\sum_{i=1}^{n} e^{x_i} \right) \left(\sum_{i=1}^{n} v_i^2 e^{x_i} \right) - \left(\sum_{i=1}^{n} v_i e^{x_i} \right)^2 \right) \geq 0 \tag{17}$$

Take the Cauchy-Schwarz inequality:

$$(\boldsymbol{a^T a})(\boldsymbol{b^T b}) \geq (\boldsymbol{a^T b})^2 \quad \forall \boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^n \tag{18}$$

If $a_i = v_i \sqrt{e^{x_i}}$ and $b_i = \sqrt{e^{x_i}}$, we obtain:

$$\left(\sum_{i=1}^{n} v_i^2 e^{x_i} \right) \left(\sum_{i=1}^{n} e^{x_i} \right) \geq \left(\sum_{i=1}^{n} v_i e^{x_i} \right)^2 \tag{19}$$

$$\left(\sum_{i=1}^{n} v_i^2 e^{x_i} \right) \left(\sum_{i=1}^{n} e^{x_i} \right) - \left(\sum_{i=1}^{n} v_i e^{x_i} \right)^2 \geq 0 \tag{20}$$

This last equation is the inside of $v^T \nabla^2 f(\boldsymbol{x}) v$. Therefore f is convex.

Geometric Mean (Bonus (0.5 pt))

Show that the geometric mean

$$f(\boldsymbol{x}) = \left(\prod_{i=1}^{n} x_i \right)^{1/n}$$

is concave on $\mathbb{R}^n_{++}$.

Solution: One possible way is to analyze the Hessian (Boyd's book p. 74) but there is a more elegant solution.

We will use the AM-GM inequality: The arithmetic mean is greater or equal than the geometric mean.

$$\frac{1}{n} \sum_{i=1}^{n} a_i \geq \left(\prod_{i=1}^{n} a_i \right)^{1/n}$$

Replace in the previous equation the following two values:

$$a_i = \frac{x_i}{\lambda x_i + (1 - \lambda) y_i} \quad b_i = \frac{y_i}{\lambda x_i + (1 - \lambda) y_i}$$

To obtain two inequalities:

$$\left(\prod_{i=1}^{n} \frac{x_i}{\lambda x_i + (1-\lambda) y_i}\right)^{1/n} = \frac{f(x_i)}{f(\lambda x_i + (1-\lambda) y_i)} \leq \frac{1}{n} \sum_{i=1}^{n} \frac{x_i}{\lambda x_i + (1-\lambda) y_i} \tag{21}$$

$$\left(\prod_{i=1}^{n} \frac{y_i}{\lambda x_i + (1-\lambda) y_i}\right)^{1/n} = \frac{f(y_i)}{f(\lambda x_i + (1-\lambda) y_i)} \leq \frac{1}{n} \sum_{i=1}^{n} \frac{y_i}{\lambda x_i + (1-\lambda) y_i} \tag{22}$$

Now multiply (21) by λ and (22) by $(1-\lambda)$:

$$\frac{\lambda f(x_i)}{f(\lambda x_i + (1-\lambda) y_i)} \leq \frac{1}{n} \sum_{i=1}^{n} \frac{\lambda x_i}{\lambda x_i + (1-\lambda) y_i} \tag{23}$$

$$\frac{(1-\lambda) f(y_i)}{f(\lambda x_i + (1-\lambda) y_i)} \leq \frac{1}{n} \sum_{i=1}^{n} \frac{(1-\lambda) y_i}{\lambda x_i + (1-\lambda) y_i} \tag{24}$$

Now sum both (23) and (24):

$$\frac{\lambda f(x_i) + (1-\lambda) f(y_i)}{f(\lambda x_i + (1-\lambda) y_i)} \leq \frac{1}{n} \sum_{i=1}^{n} \left(\frac{\lambda x_i}{\lambda x_i + (1-\lambda) y_i} + \frac{(1-\lambda) y_i}{\lambda x_i + (1-\lambda) y_i}\right) = 1$$

$$\frac{\lambda f(x_i) + (1-\lambda) f(y_i)}{f(\lambda x_i + (1-\lambda) y_i)} \leq 1$$

$$\Longrightarrow \lambda f(x_i) + (1-\lambda) f(y_i) \leq f(\lambda x_i + (1-\lambda) y_i)$$

Now take $g(\boldsymbol{x}) = -f(\boldsymbol{x})$ and replace:

$$g(\lambda x_i + (1-\lambda) y_i) \leq \lambda g(x_i) + (1-\lambda) g(y_i)$$

Which is the definition of a convex function. Since $g(\boldsymbol{x}) = -f(\boldsymbol{x})$, then $f(\boldsymbol{x})$ is a concave function.